

Solução do Problema Sc205 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: Sc205
- Número de Variáveis: 203
- Número de Restrições: 205
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 0.000e+00
- Valor Primal: -5.220e+01
- Valor Dual: -5.220e+01
- Gap: -1.033e-17
- Número de Iterações: 12

2 Variáveis Primais e Custos Reduzidos

Tabela 1: Variáveis primais (x) e custos reduzidos (z) do problema Sc205

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	0.000e+00	1	0.000e+00
2	1.085e+01	2	0.000e+00
3	5.220e+01	3	0.000e+00
4	5.220e+01	4	0.000e+00
5	5.220e+01	5	0.000e+00
6	0.000e+00	6	0.000e+00
7	1.085e+01	7	0.000e+00
8	5.220e+01	8	0.000e+00
9	0.000e+00	9	1.500e-01
10	1.085e+01	10	0.000e+00
11	5.220e+01	11	0.000e+00
12	0.000e+00	12	9.516e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
13	1.236e+01	13	0.000e+00
14	5.742e+01	14	0.000e+00
15	5.742e+01	15	0.000e+00
16	1.096e+02	16	0.000e+00
17	0.000e+00	17	0.000e+00
18	2.321e+01	18	0.000e+00
19	1.096e+02	19	0.000e+00
20	0.000e+00	20	0.000e+00
21	2.321e+01	21	0.000e+00
22	1.096e+02	22	0.000e+00
23	0.000e+00	23	5.863e-02
24	1.423e+01	24	0.000e+00
25	6.316e+01	25	0.000e+00
26	6.316e+01	26	0.000e+00
27	1.728e+02	27	0.000e+00
28	0.000e+00	28	0.000e+00
29	3.744e+01	29	0.000e+00
30	1.728e+02	30	0.000e+00
31	0.000e+00	31	0.000e+00
32	3.744e+01	32	0.000e+00
33	1.728e+02	33	0.000e+00
34	0.000e+00	34	3.428e-02
35	1.661e+01	35	0.000e+00
36	6.948e+01	36	0.000e+00
37	6.948e+01	37	0.000e+00
38	2.423e+02	38	0.000e+00
39	0.000e+00	39	0.000e+00
40	5.404e+01	40	0.000e+00
41	2.423e+02	41	0.000e+00
42	0.000e+00	42	0.000e+00
43	5.404e+01	43	0.000e+00
44	2.423e+02	44	0.000e+00
45	0.000e+00	45	1.804e-02
46	1.970e+01	46	0.000e+00
47	7.643e+01	47	0.000e+00
48	7.643e+01	48	0.000e+00
49	3.187e+02	49	0.000e+00
50	0.000e+00	50	0.000e+00
51	7.374e+01	51	0.000e+00
52	3.187e+02	52	0.000e+00
53	0.000e+00	53	0.000e+00
54	7.374e+01	54	0.000e+00
55	3.187e+02	55	0.000e+00
56	0.000e+00	56	7.216e-03
57	2.382e+01	57	0.000e+00
58	8.407e+01	58	0.000e+00
59	8.407e+01	59	0.000e+00
60	4.028e+02	60	0.000e+00
61	0.000e+00	61	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
62	9.756e+01	62	0.000e+00
63	4.028e+02	63	0.000e+00
64	0.000e+00	64	0.000e+00
65	9.756e+01	65	0.000e+00
66	4.028e+02	66	0.000e+00
67	1.799e+01	67	0.000e+00
68	2.043e+01	68	0.000e+00
69	9.248e+01	69	0.000e+00
70	9.248e+01	70	0.000e+00
71	4.952e+02	71	0.000e+00
72	1.799e+01	72	0.000e+00
73	1.180e+02	73	0.000e+00
74	4.952e+02	74	0.000e+00
75	1.799e+01	75	0.000e+00
76	1.180e+02	76	0.000e+00
77	4.952e+02	77	0.000e+00
78	2.180e+01	78	0.000e+00
79	2.180e+01	79	0.000e+00
80	1.017e+02	80	0.000e+00
81	1.017e+02	81	0.000e+00
82	5.970e+02	82	0.000e+00
83	3.979e+01	83	0.000e+00
84	1.398e+02	84	0.000e+00
85	5.970e+02	85	0.000e+00
86	3.979e+01	86	0.000e+00
87	1.398e+02	87	0.000e+00
88	5.970e+02	88	0.000e+00
89	2.398e+01	89	0.000e+00
90	2.398e+01	90	0.000e+00
91	1.119e+02	91	0.000e+00
92	1.119e+02	92	0.000e+00
93	7.089e+02	93	0.000e+00
94	6.376e+01	94	0.000e+00
95	1.638e+02	95	0.000e+00
96	7.089e+02	96	0.000e+00
97	6.376e+01	97	0.000e+00
98	1.638e+02	98	0.000e+00
99	7.089e+02	99	0.000e+00
100	2.638e+01	100	0.000e+00
101	2.638e+01	101	0.000e+00
102	1.231e+02	102	0.000e+00
103	0.000e+00	103	0.000e+00
104	1.231e+02	104	0.000e+00
105	8.320e+02	105	0.000e+00
106	9.014e+01	106	0.000e+00
107	1.901e+02	107	0.000e+00
108	8.320e+02	108	0.000e+00
109	9.014e+01	109	0.000e+00
110	1.901e+02	110	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
111	8.320e+02	111	0.000e+00
112	2.901e+01	112	0.000e+00
113	2.901e+01	113	0.000e+00
114	1.354e+02	114	0.000e+00
115	1.354e+02	115	0.000e+00
116	9.674e+02	116	0.000e+00
117	1.192e+02	117	0.000e+00
118	2.192e+02	118	0.000e+00
119	9.674e+02	119	0.000e+00
120	1.192e+02	120	0.000e+00
121	2.192e+02	121	0.000e+00
122	9.674e+02	122	0.000e+00
123	3.192e+01	123	0.000e+00
124	3.192e+01	124	0.000e+00
125	1.489e+02	125	0.000e+00
126	1.489e+02	126	0.000e+00
127	1.116e+03	127	0.000e+00
128	1.511e+02	128	0.000e+00
129	2.511e+02	129	0.000e+00
130	1.116e+03	130	0.000e+00
131	1.511e+02	131	0.000e+00
132	2.511e+02	132	0.000e+00
133	1.116e+03	133	0.000e+00
134	3.511e+01	134	0.000e+00
135	3.511e+01	135	0.000e+00
136	1.638e+02	136	0.000e+00
137	1.638e+02	137	0.000e+00
138	1.280e+03	138	0.000e+00
139	1.862e+02	139	0.000e+00
140	2.862e+02	140	0.000e+00
141	1.280e+03	141	0.000e+00
142	1.862e+02	142	0.000e+00
143	2.862e+02	143	0.000e+00
144	1.280e+03	144	0.000e+00
145	3.862e+01	145	0.000e+00
146	3.862e+01	146	0.000e+00
147	1.802e+02	147	0.000e+00
148	1.802e+02	148	0.000e+00
149	1.460e+03	149	0.000e+00
150	2.248e+02	150	0.000e+00
151	3.248e+02	151	0.000e+00
152	1.460e+03	152	0.000e+00
153	2.248e+02	153	0.000e+00
154	3.248e+02	154	0.000e+00
155	1.460e+03	155	0.000e+00
156	4.248e+01	156	0.000e+00
157	4.248e+01	157	0.000e+00
158	1.982e+02	158	0.000e+00
159	1.982e+02	159	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
160	1.659e+03	160	0.000e+00
161	2.673e+02	161	0.000e+00
162	3.673e+02	162	0.000e+00
163	1.659e+03	163	0.000e+00
164	2.673e+02	164	0.000e+00
165	3.673e+02	165	0.000e+00
166	1.659e+03	166	0.000e+00
167	4.673e+01	167	0.000e+00
168	4.673e+01	168	0.000e+00
169	2.181e+02	169	0.000e+00
170	2.181e+02	170	0.000e+00
171	1.877e+03	171	0.000e+00
172	3.140e+02	172	0.000e+00
173	4.140e+02	173	0.000e+00
174	1.877e+03	174	0.000e+00
175	3.140e+02	175	0.000e+00
176	4.140e+02	176	0.000e+00
177	1.877e+03	177	0.000e+00
178	5.140e+01	178	0.000e+00
179	5.140e+01	179	0.000e+00
180	2.399e+02	180	0.000e+00
181	2.399e+02	181	0.000e+00
182	2.117e+03	182	0.000e+00
183	3.654e+02	183	0.000e+00
184	4.654e+02	184	0.000e+00
185	2.117e+03	185	0.000e+00
186	3.654e+02	186	0.000e+00
187	4.654e+02	187	0.000e+00
188	2.117e+03	188	0.000e+00
189	5.654e+01	189	0.000e+00
190	5.654e+01	190	0.000e+00
191	2.639e+02	191	0.000e+00
192	2.639e+02	192	0.000e+00
193	2.380e+03	193	0.000e+00
194	4.219e+02	194	0.000e+00
195	5.219e+02	195	0.000e+00
196	2.380e+03	196	0.000e+00
197	4.219e+02	197	0.000e+00
198	5.219e+02	198	0.000e+00
199	2.380e+03	199	0.000e+00
200	6.219e+01	200	0.000e+00
201	6.219e+01	201	0.000e+00
202	2.902e+02	202	0.000e+00
203	2.902e+02	203	0.000e+00

3 Variáveis Duais (Multiplicadores de Lagrange)

Tabela 2: Variáveis duais (y) do problema Sc205

Coordenada y	Valor y (Dual)
1	0.000e+00
2	-1.644e-01
3	0.000e+00
4	-2.466e-01
5	-1.644e-01
6	-3.288e-01
7	-2.466e-01
8	-1.500e-01
9	-1.096e-01
10	-8.219e-02
11	0.000e+00
12	-1.096e-01
13	-8.219e-02
14	-6.849e-01
15	-1.644e-01
16	-1.443e-02
17	-2.192e-01
18	-1.644e-01
19	0.000e+00
20	-7.306e-02
21	-5.480e-02
22	0.000e+00
23	-7.306e-02
24	-5.480e-02
25	-4.732e-01
26	-1.096e-01
27	-1.443e-02
28	-1.461e-01
29	-1.096e-01
30	0.000e+00
31	-4.871e-02
32	-3.653e-02
33	0.000e+00
34	-4.871e-02
35	-3.653e-02
36	-3.306e-01
37	-7.306e-02
38	-1.443e-02
39	-9.741e-02
40	-7.306e-02
41	0.000e+00
42	-3.247e-02
43	-2.435e-02
44	0.000e+00
45	-3.247e-02
46	-2.435e-02
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Coordenada y	Valor y (Dual)
47	-2.341e-01
48	-4.871e-02
49	-1.443e-02
50	-6.494e-02
51	-4.871e-02
52	0.000e+00
53	-2.165e-02
54	-1.624e-02
55	0.000e+00
56	-2.165e-02
57	-1.624e-02
58	-1.685e-01
59	-3.247e-02
60	-1.443e-02
61	-4.330e-02
62	-3.247e-02
63	0.000e+00
64	-1.443e-02
65	-1.082e-02
66	0.000e+00
67	-1.443e-02
68	-1.082e-02
69	-1.237e-01
70	-2.165e-02
71	-1.443e-02
72	-2.886e-02
73	-2.165e-02
74	-1.804e-03
75	-9.020e-03
76	-5.412e-03
77	-1.804e-03
78	-9.020e-03
79	-5.412e-03
80	-9.278e-02
81	-1.624e-02
82	-1.263e-02
83	-1.984e-02
84	-1.624e-02
85	-2.255e-03
86	-5.863e-03
87	-4.059e-03
88	-2.255e-03
89	-5.863e-03
90	-4.059e-03
91	-6.958e-02
92	-1.218e-02
93	-1.037e-02
94	-1.398e-02
95	-1.218e-02

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Coordenada y	Valor y (Dual)
96	-2.142e-03
97	-3.946e-03
98	-3.044e-03
99	-2.142e-03
100	-3.946e-03
101	-3.044e-03
102	0.000e+00
103	-5.219e-02
104	-9.133e-03
105	-8.231e-03
106	-1.003e-02
107	-9.133e-03
108	-1.832e-03
109	-2.734e-03
110	-2.283e-03
111	-1.832e-03
112	-2.734e-03
113	-2.283e-03
114	-3.914e-02
115	-6.849e-03
116	-6.398e-03
117	-7.300e-03
118	-6.849e-03
119	-1.487e-03
120	-1.938e-03
121	-1.712e-03
122	-1.487e-03
123	-1.938e-03
124	-1.712e-03
125	-2.935e-02
126	-5.137e-03
127	-4.912e-03
128	-5.363e-03
129	-5.137e-03
130	-1.172e-03
131	-1.397e-03
132	-1.284e-03
133	-1.172e-03
134	-1.397e-03
135	-1.284e-03
136	-2.202e-02
137	-3.853e-03
138	-3.740e-03
139	-3.966e-03
140	-3.853e-03
141	-9.068e-04
142	-1.020e-03
143	-9.632e-04
144	-9.068e-04

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Coordenada y	Valor y (Dual)
145	-1.020e-03
146	-9.632e-04
147	-1.651e-02
148	-2.890e-03
149	-2.833e-03
150	-2.946e-03
151	-2.890e-03
152	-6.942e-04
153	-7.506e-04
154	-7.224e-04
155	-6.942e-04
156	-7.506e-04
157	-7.224e-04
158	-1.238e-02
159	-2.167e-03
160	-2.139e-03
161	-2.195e-03
162	-2.167e-03
163	-5.277e-04
164	-5.559e-04
165	-5.418e-04
166	-5.277e-04
167	-5.559e-04
168	-5.418e-04
169	-9.288e-03
170	-1.625e-03
171	-1.611e-03
172	-1.639e-03
173	-1.625e-03
174	-3.993e-04
175	-4.134e-04
176	-4.063e-04
177	-3.993e-04
178	-4.134e-04
179	-4.063e-04
180	-6.966e-03
181	-1.219e-03
182	-1.212e-03
183	-1.226e-03
184	-1.219e-03
185	-3.012e-04
186	-3.083e-04
187	-3.048e-04
188	-3.012e-04
189	-3.083e-04
190	-3.048e-04
191	-5.224e-03
192	-9.143e-04
193	-9.108e-04

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Coordenada y	Valor y (Dual)
194	-9.178e-04
195	-9.143e-04
196	-9.108e-04
197	-9.178e-04
198	-9.143e-04
199	-9.108e-04
200	-9.178e-04
201	-9.143e-04
202	-3.918e-03
203	-3.914e-03
204	-3.922e-03
205	-3.918e-03